Quarter-Car Suspension Damping Analysis

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1. Project Summary

I used a passive quarter-car model in MATLAB to investigate how damping and sprung mass affect the suspension response to a road bump. I compared five damping coefficients at vehicle speeds of 15 m/s and 20 m/s, using an assumed bump 25 mm high and 3 m long. Each setting was assessed using root mean square (RMS) tyre-load variation, RMS body acceleration and peak suspension travel. I then increased the sprung mass by 10%, from 250 kg to 275 kg, to check whether the preferred damping setting changed.

A damping coefficient of 1,000 N·s/m gave the lowest RMS tyre-load variation among the tested settings at both speeds and both masses. This setting also remained within the assumed 40 mm dynamic suspension-travel limit and maintained positive modelled tyre load. Increasing damping above this value reduced peak suspension travel but increased RMS body acceleration and tyre-load variation. These findings apply to the assumed model and tested bump conditions. The model has not been validated against physical test data.

2. Model and assumptions

I used a passive two-degree-of-freedom quarter-car model to investigate the vertical response of a vehicle travelling over a road bump. The model consists of a sprung mass, representing the portion of the vehicle body supported by the suspension, and an unsprung mass, representing the wheel assembly. The two masses are connected by a suspension spring and damper, while the tyre is represented by a spring connecting the wheel to the road. I used the equations from the MathWorks example [1], with the active suspension force set to zero. The vehicle and road parameters were assumed for this study.

Table 1. Model Parameters and Simulation Conditions

Parameter	Symbol	Value(s) tested / used	Unit
Sprung mass - original test	m_s	250	kg
Sprung mass - increased mass test	m_s	275	kg
Unsprung mass	m_u	35	kg
Suspension spring stiffness	k_s	18,000	N/m
Tyre stiffness	k_t	180,000	N/m
Damping coefficients tested	c	500, 1,000, 1,500, 2,000, 3,000	N·s/m
Bump shape	—	Raised cosine	—
Bump height	h	25	mm
Bump length	L	3	m
Vehicle speeds tested	v	15, 20	m/s
Assumed dynamic suspension travel limit	—	40	mm

2.1 Equations of Motion

The vertical motion of the sprung and unsprung masses was calculated using two coupled equations of motion. These equations describe how the suspension spring, damper and tyre stiffness affect the vehicle's response when travelling over a road bump.

Equation 1: Sprung Mass

$$m_s\ddot{z}_s = -k_s(z_s - z_u) - c(\dot{z}_s - \dot{z}_u)$$

Equation 2: Unsprung Mass

$$m_u\ddot{z}_u = k_s(z_s - z_u) + c(\dot{z}_s - \dot{z}_u) - k_t(z_u - z_r)$$

Displacements are measured from the loaded static equilibrium position, with upward movement taken as positive. Here, z_s represents the vertical displacement of the vehicle body, z_u represents the wheel displacement, and z_r represents the road input. A single dot above a variable indicates velocity, while two dots indicate acceleration.

The relative displacement between the sprung and unsprung masses (z_s − z_u) represents the dynamic suspension travel. This was used to check whether the suspension remained within the assumed 40 mm travel limit.

2.2 Tyre-load Calculation

I calculated the vertical tyre load using the tyre stiffness and the relative displacement between the road and wheel. This allowed me to investigate how changes in damping affected the load acting through the tyre during the bump.

The static tyre load was calculated from the combined sprung and unsprung mass at the modelled corner. I then used the variation from this static load to compare the different damping coefficients.

Total Vertical Tyre Load:

$$N(t) = N_0 + k_t(z_r - z_u)$$

Static Tyre Load:

$$N_0 = (m_s + m_u)g$$

Tyre load Variation:

$$\Delta N(t) = N(t) - N_0$$

3. Simulation and selection method

I used MATLAB's ode45 solver [2] to simulate each case over 3 seconds, with results recorded at 1 ms intervals. The bump began at 0.5 s and lasted 0.20 s at 15 m/s or 0.15 s at 20 m/s.

To compare the damping coefficients, I calculated the RMS tyre-load variation over the same 0.5–3 s period, expressed as a percentage of the static tyre load. I also checked that the maximum dynamic suspension travel remained within the assumed 40 mm limit and that the modelled tyre load remained positive.

I then increased the sprung mass from 250 kg to 275 kg, keeping the other input parameters unchanged, to investigate how additional vehicle mass affected the results.

4. Hand estimates and numerical checks

I carried out hand calculations to check the approximate natural frequency, damping ratio and static tyre load of the model. The frequency and damping calculations assume the wheel remains fixed, providing an estimate of the vehicle body's response rather than the exact behaviour of the full two-degree-of-freedom system.

$$f_n \approx \frac{1}{2\pi}\sqrt{\frac{k_s}{m_s}}$$

$$c_{\mathrm{cr}} \approx 2\sqrt{k_s m_s}$$

$$\zeta \approx \frac{c}{c_{\mathrm{cr}}}$$

$$N_0 = (m_s + m_u)g$$

f_n is the approximate body natural frequency, c_{cr} is the critical damping coefficient, ζ is the approximate damping ratio, N_0 and is the static tyre load at the modelled corner.

The damping ratios were calculated using a damping coefficient of 1,000 N·s/m. The hand-calculated values are summarised in Table 2.

Table 2 Hand Calculated Estimates for the two sprung masses

Check	250kg	275kg
Approximate body frequency	1.350 Hz	1.288 Hz
Approximate damping ratio	0.236	0.225
Static corner load	2795.85 N	3041.10 N

I carried out numerical checks to assess the consistency of the MATLAB results. With no road input and zero initial displacement and velocity, the model remained at its static equilibrium position, as expected.

I then repeated selected simulations using a smaller output interval of 0.5 ms, tighter solver tolerances and half the original maximum solver step. The largest relative changes in the compared results were 0.0037% for the 250 kg case and 0.0117% for the 275 kg case, both below the 1% threshold selected for this study.

These small differences suggest that the results were not significantly affected by the numerical settings tested. However, these checks do not replace validation against physical test data.

5. Results and Damping Selection

I compared 20 simulation cases covering five damping coefficients, two vehicle speeds and two sprung masses. All cases remained within the assumed 40 mm suspension-travel limit and maintained a positive modelled tyre load.
A damping coefficient of 1,000 N·s/m produced the lowest RMS tyre-load variation for both sprung masses at both vehicle speeds. I therefore selected this coefficient as the preferred setting from the values tested. The results for this setting are summarised in Table 3.

Table 3 Simulation Results at the Selected Damping Coefficient (1000 N.s/m)

Sprung mass (kg)	Speed (m/s)	Tyre-Load RMS (%)	Body Acceleration RMS (m/s^2)	Peak Suspension Travel (mm)	Min. Tyre Load (N)
250	15	5.566	0.568	21.19	2236
250	20	7.890	0.622	24.73	1565
275	15	5.085	0.517	21.50	2504
275	20	7.208	0.564	25.00	1834

The results show that increasing vehicle speed from 15 m/s to 20 m/s increased the RMS tyre-load variation, body acceleration and peak suspension travel for both sprung masses. For the 250 kg model, RMS tyre-load variation increased from 5.566% to 7.890%, while peak suspension travel increased from 21.19 mm to 24.73 mm.
Increasing the sprung mass to 275 kg reduced the normalised RMS tyre-load variation and body acceleration at both speeds. However, peak suspension travel increased slightly. This shows how changing the vehicle mass affects the suspension response, even when the spring stiffness and damping coefficient remain unchanged.

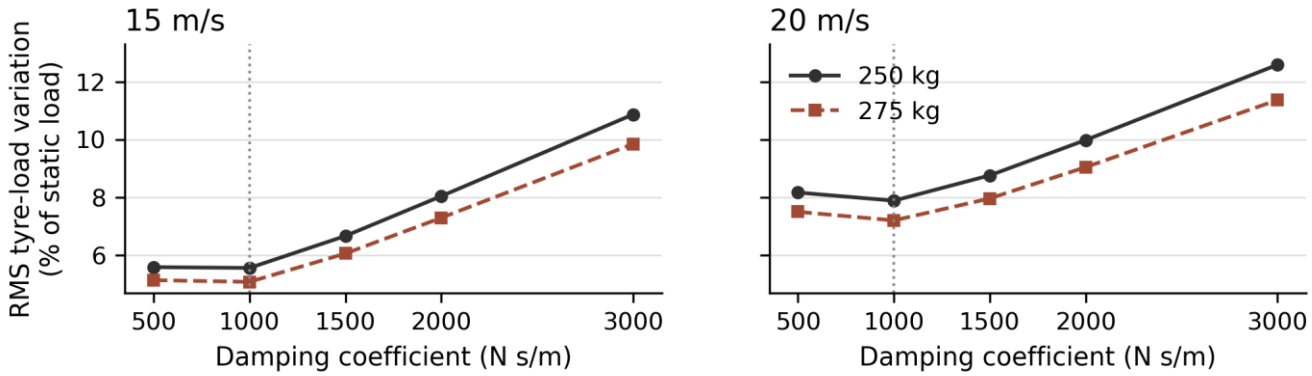


Figure 1. RMS tyre-load variation against damping coefficient at 15 m/s and 20 m/s. The dotted line indicates the selected coefficient of 1,000 N·s/m. Lines connect the five discrete test values and do not establish a continuous optimum.

Figure 1 shows that RMS tyre-load variation was lowest at 1,000 N·s/m across the four mass and speed combinations. Increasing the damping coefficient beyond this value led to a greater variation in tyre load, particularly at 2,000 and 3,000 N·s/m.

Although 1,000 N·s/m gave the best results for the selected tyre-load criterion, this does not mean it would necessarily provide the best overall ride comfort or handling performance. The preferred setting is based on the performance criteria and operating conditions used in this study.

6. Damping Trade-Off and Mass Sensitivity

Increasing the damping coefficient above 1,000 N·s/m reduced peak suspension travel but increased tyre-load variation and body acceleration. Although 1,000 N·s/m produced the lowest RMS tyre-load variation, the 500 N·s/m setting gave the lowest body acceleration. For the 250 kg model at 15 m/s, tyre-load variation at 500 N·s/m was only 0.49% higher than at 1,000 N·s/m which represents a relative percentage increase. This highlights the trade-off between tyre-load variation, suspension travel and ride comfort, with body acceleration used as an indicator of ride comfort.

I also investigated how increasing the sprung mass from 250 kg to 275 kg affected the results. At 1,000 N·s/m, the normalised RMS tyre-load variation decreased by approximately 8.65% at both speeds. However, much of this reduction was due to the higher static tyre load used in the percentage calculation. At 20 m/s, the actual RMS tyre-load variation decreased only slightly, from 220.60 N to 219.19 N. Body acceleration also decreased, although peak suspension travel increased slightly. Despite these changes, 1,000 N·s/m remained the preferred damping coefficient for both sprung masses under the selected tyre-load criterion.

7. Recommendations and Limits

Based on the simulation results, I would recommend a damping coefficient of 1,000 N·s/m for the conditions tested, as it produced the lowest RMS tyre-load variation while remaining within the assumed suspension-travel limit. However, this was the preferred value from five tested coefficients rather than an overall optimum. Further testing with different road profiles and damping values would help determine whether this setting remains suitable under other conditions.

The model also has limitations. It assumes linear spring, damper and tyre behaviour and does not account for vehicle pitch, roll or aerodynamic effects. As the study was based entirely on simulation, the results cannot be used to confirm improvements in real-world grip or lap times. Future work could involve using measured vehicle parameters and comparing the simulation results with physical testing to improve the reliability of the model.

8. References and saved evidence

[1] MathWorks, *Control Quarter-Car Suspension Dynamics Using ADMM Solver*. MATLAB & Simulink documentation. Available: https://www.mathworks.com/help/mpc/ug/admm-based-mpc-control-for-quarter-car-suspension.html

The passive quarter-car equations were adapted from this example. The active suspension controller was not used.

[2] MathWorks, *ode45 — Solve Nonstiff Differential Equations*. MATLAB documentation. Available: https://www.mathworks.com/help/matlab/ref/ode45.html

Supporting Files

The original and increased-mass MATLAB simulation runs, dated 11 October 2026, are retained alongside the project. These include comparison tables, numerical checks, response plots, run notes and saved MATLAB data used to assess the different damping coefficients.